

This result implies that in general, $\beta_{fe} \neq \delta^{TR}$, so $\hat{\beta}_{fe}$ is a biased estimator of the ATT. To illustrate this, we consider a simple example of a staggered adoption design with two groups and three periods, and where the treatments are non-stochastic: group 1 is untreated at periods 1 and 2 and treated at period 3, while group 2 is untreated at period 1 and treated both at periods 2 and 3.⁸ We also assume that $N_{g,t}/N_{g,t-1}$ does not vary across g : all groups experience the same growth of their number of observations from $t - 1$ to t , a requirement that is for instance satisfied when the data is a balanced panel. Then, one can show that

$$\varepsilon_{g,t} = D_{g,t} - D_{g,\cdot} - D_{\cdot,t} + D_{\cdot,\cdot},$$

thus implying that

$$\begin{aligned}\varepsilon_{1,3} &= 1 - 1/3 - 1 + 1/2 = 1/6, \\ \varepsilon_{2,2} &= 1 - 2/3 - 1/2 + 1/2 = 1/3, \\ \varepsilon_{2,3} &= 1 - 2/3 - 1 + 1/2 = -1/6.\end{aligned}$$

The residual is negative in group 2 and period 3, because the regression predicts a treatment probability larger than one in that cell, a classic extrapolation problem with linear regressions. Then, under the common trends assumption, it follows from Theorem 1 and the fact that the treatments are non-stochastic that

$$\beta_{fe} = 1/2E[\Delta_{1,3}] + E[\Delta_{2,2}] - 1/2E[\Delta_{2,3}].$$

β_{fe} is equal to a weighted sum of the ATEs in group 1 at period 3, group 2 at period 2, and group 2 at period 3, the three treated (g, t) cells. However, the weight assigned to each ATE differs from $1/3$, the proportion that each cell accounts for in the population of treated observations. Therefore, β_{fe} is not equal to δ^{TR} . Perhaps more worryingly, not all the weights are positive: the weight assigned to the ATE in group 2 period 3 is strictly negative. Consequently, β_{fe} may be a very misleading measure of the treatment effect. Assume for instance that $E[\Delta_{1,3}] = E[\Delta_{2,2}] = 1$ and $E[\Delta_{2,3}] = 4$. At the period when they start receiving the treatment, both groups experience a modest positive ATE. But this effect builds over time and in period 3, one period after it has started receiving the treatment, group 2 now experiences a large ATE. Then,

$$\beta_{fe} = 1/2 \times 1 + 1 - 1/2 \times 4 = -1/2.$$

β_{fe} is strictly negative, while $E[\Delta_{1,3}]$, $E[\Delta_{2,2}]$, and $E[\Delta_{2,3}]$ are all positive. More generally, the negative weights are an issue if the $E[\Delta_{g,t}]$ s are heterogeneous, across groups or over time.⁹ If $E[\Delta_{1,3}] = E[\Delta_{2,2}] = E[\Delta_{2,3}] = 1$, then $\beta_{fe} = 1 = \delta^{TR}$.

⁸ A similar example appears in Borusyak and Jaravel (2017).

⁹ On the other hand, β_{fe} does not rule out heterogeneous treatment effects within (g, t) cells, as it is identified by variations across (g, t) cells, and does not leverage any within-cell variation.